

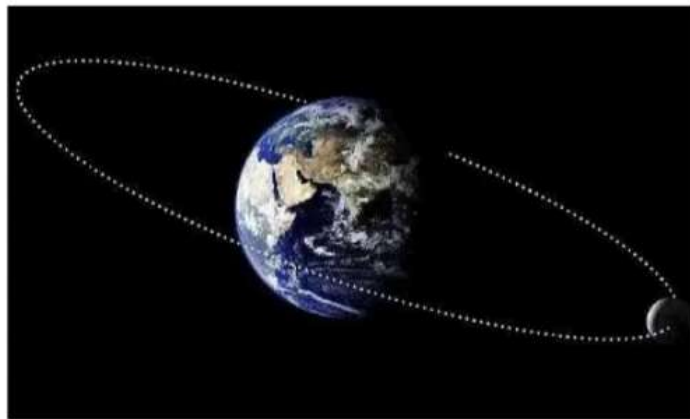
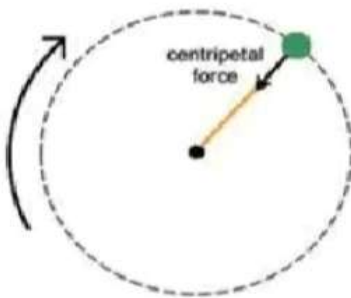
Gravitation

Circular motion: Circular motion is a type of motion in which an object moves around a fixed point or axis, following a circular path. This type of motion is characterized by a constant radius from the centre of the circle and a continuous change in direction.

Key characteristics of circular motion include:

****Centripetal Force:**** Circular motion requires a force directed toward the center of the circle to keep the object moving in that path. This force is called the centripetal force, and it is necessary to counteract the tendency of an object to move in a straight line due to its inertia.

****Examples:**** Examples of circular motion include the Moon

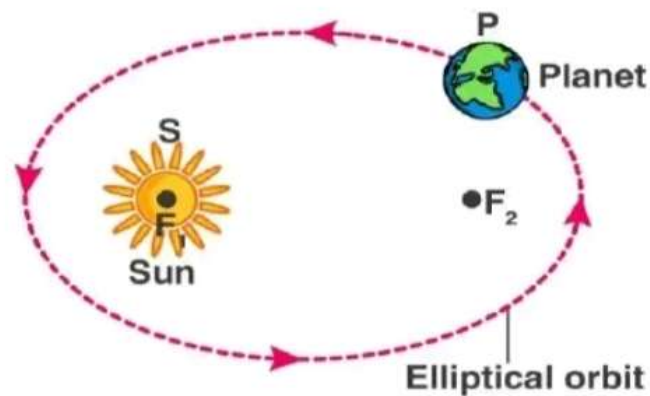


orbiting the Earth, a car moving around a curve, a spinning top, or a ball on a string being twirled around.

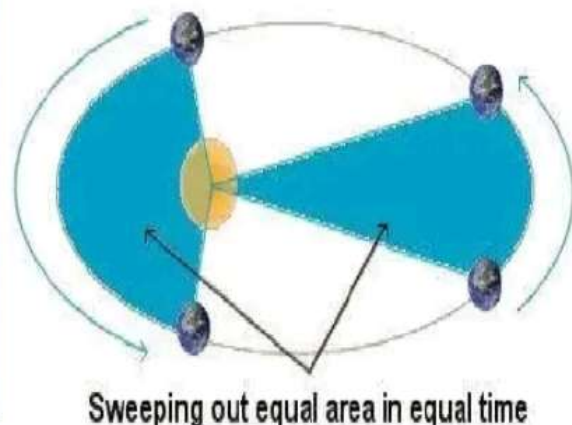
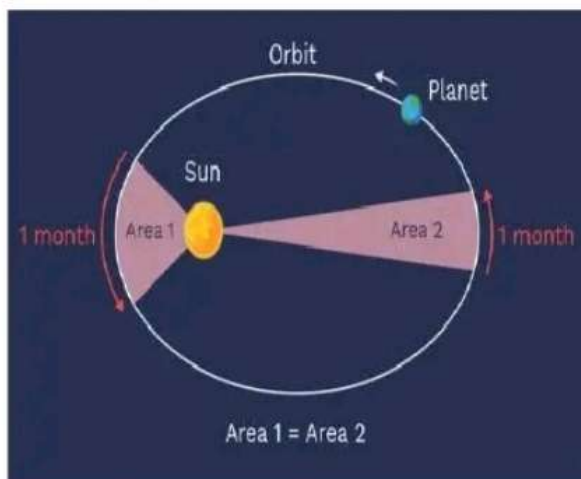
Q. Explain Kepler's laws of planetary motion

Johannes Kepler's Laws of Planetary Motion, formulated in the early 17th century, are fundamental principles describing the motion of celestial bodies, particularly planets in our solar system. There are three laws:

1. ****Kepler's First Law (Law of Ellipses):**** The orbits of planets are **ellipses** with the Sun at one of the two **foci**. **An ellipse is a stretched-out circle.** The Sun is located at one of the two fixed points within the ellipse called the foci.



2. ****Kepler's Second Law (Law of Equal Areas):**** This law states that a line segment joining a planet and the Sun **sweeps out equal areas in equal amounts of time**. When a planet is closer to the Sun, it moves faster in its orbit (as shown in the diagram below).



3. **Kepler's Third Law (Law of Harmonies):** The square of the orbital period (T) of a planet is directly proportional to the cube of the semi-major axis (r) of its orbit. This can be expressed mathematically as:

$$T^2 / r^3 = \text{constant}$$

i.e.

T² is directly proportional to r³

Here, T is the time it takes for a planet to complete one orbit (in years), and a is the semi-major axis of the planet's elliptical orbit (the average distance from the Sun to the planet).

Q. Write about Newton's law of gravitation:

Newton's Law of Gravitation, formulated by Sir Isaac Newton in the late 17th century, is a fundamental principle in physics that describes the force of gravity between two objects. The law is expressed as follows:

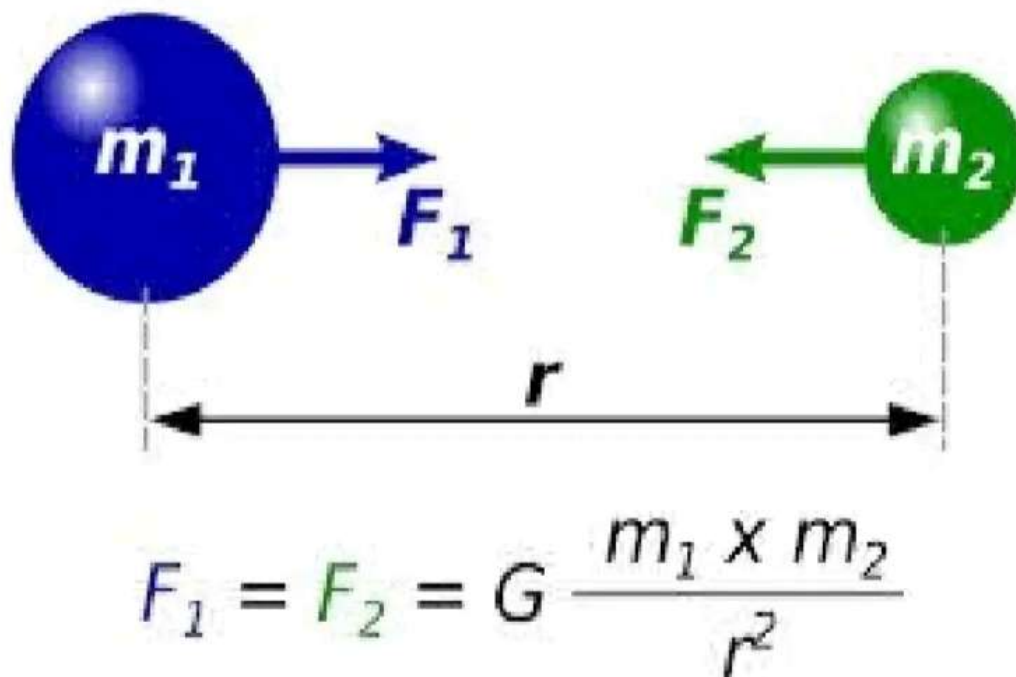
*****"Every point mass attracts every other point mass by a force acting along the line intersecting both points. This force is directly proportional to the product of the masses of the objects and inversely proportional to the square of the distance between them."****

Mathematically, Newton's Law of Gravitation is represented as:

$$F = Gm_1m_2 / r^2$$

Where:

- F is the force of gravity between two objects.
- G is the gravitational constant, a universal constant that determines the strength of the gravitational force.
- m₁ and m₂ are the masses of the two objects.
- r is the distance between the centers of the two objects.



****Earth' gravitational force****The earth attracts every object near it towards itself because of the gravitational force. The centre of mass of the earth is situated at its centre, so the gravitational force on any object .Due to the earth is always directed towards the centre of the earth. Because of this force, an Object falls vertically downwards on the earth.

Q. Explain the variation in the value of acceleration due to gravity(g):

A. Change along the surface of the earth : Will the value of g be the same everywhere on The surface of the earth? The answer is no. The reason is that the shape of the earth is not exactly spherical and so the distance of a point on the surface of the earth from its centre differs somewhat from place to place. Due to its rotation, the earth bulges at the equator and is flatter at the poles. Its radius is largest at the equator and smallest at the poles. **The value of g is thus Highest (9.832 m/s²) at the poles and decreases slowly with decreasing latitude. It is lowest (9.78 m/s²) at the equator.**

B. Change with height : **As we go above the earth's surface, the radius Increases and the value of g decreases.** However, the decrease is rather small for heights Which are small in comparison to the earth's radius. For example, remember that the radius Of the earth is 6400 km. If an aeroplane is flying at a height 10 km above the surface of the Earth, its distance from the earth's surface changes from 6400 km to 6410 km and the change In the value of g due to it is negligible. On the other hand, when we consider an artificial Satellite orbiting the earth, we have to take into account the change in the value of g due to the Large change in the distance of the satellite from the centre of the earth. Some typical heights And the values of g at these heights are given in the following table.

C. Change with depth : The value of g also changes if we go inside the earth. The value Of r decreases and one would think that the value of g should increase as Per the formula. However, the part of the earth which contributes towards the gravitational force felt by the object also decreases. Which means that the value of M also

decreases. As a combined result of change in r and M , **the value of g Decreases as we go deep inside the earth.**

Q. Differentiate between mass and weight:

The relation between mass and weight is a fundamental concept in physics and is often explained in terms of the force of gravity. Here's how mass and weight are related:

****Mass:**** Mass is a measure of the amount of matter in an object.

- 1. It is a scalar quantity and is typically measured in kilograms (kg) in the International System of Units (SI).**
- 2. Mass doesn't change regardless of its location in the universe.**

****Weight:**** Weight is the force with which an object is pulled toward the center of the Earth due to gravity.

- 1. It is a vector quantity and is typically measured in newtons (N) in the SI system.**
- 2. Weight depends on both the mass of the object and the strength of the gravitational field it is in.**

The relationship between mass and weight can be described using the following equation:

$$\text{Weight (W)} = \text{Mass (m)} \times \text{Acceleration Due to Gravity (g)}$$

Gravitational potential energy:

It is the energy something has because it's high up. Imagine a ball on a shelf. The higher the shelf, the more energy the ball has. When you let the ball fall, this energy makes it move. So, the higher something is, the more energy it has when it falls down. That's gravitational potential energy.

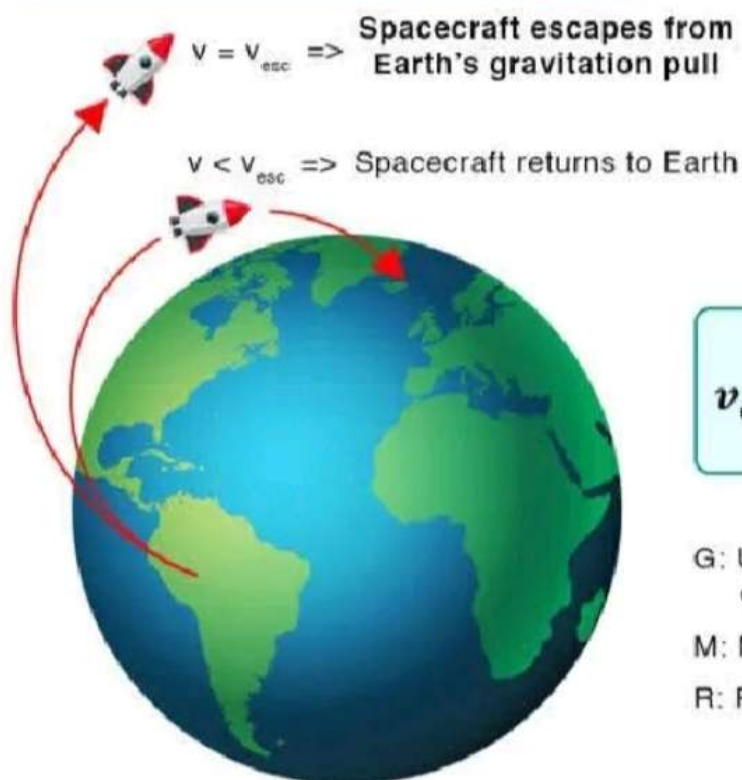
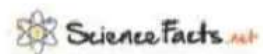
Q. What is escape velocity?

Escape velocity is the minimum speed that an object must reach to break free from the gravitational pull of a celestial body, such as a planet or moon, and move into space without falling back. In other words, it's the speed needed for an object to escape or "launch" into space, never to return to the surface.

Escape velocity depends on the mass and size of the celestial body. The larger and more massive the body, the greater the escape velocity required. For example, on Earth, the escape velocity is approximately 11.2 kilometers per second

To leave Earth and enter space, rockets must reach or exceed this speed. If they don't, they won't have enough energy to overcome Earth's gravitational pull and will either fall back or enter orbit around the planet. Escape velocity is a critical concept in space exploration and is used to plan missions to other planets and celestial objects.

Escape Velocity (v_{esc})



$v = v_{esc} \Rightarrow$ Spacecraft escapes from Earth's gravitation pull

$v < v_{esc} \Rightarrow$ Spacecraft returns to Earth

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

G: Universal gravitational constant

M: Mass of Earth

R: Radius of Earth